

# myTutor Developer Guide

Architecture, prerequisites, setup, configuration and operations

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Author: Naw Aung (81017125), UC Executive MBA · MBAM601 Digital Experiment
Repository: github.com/henrynawaung/mytutor

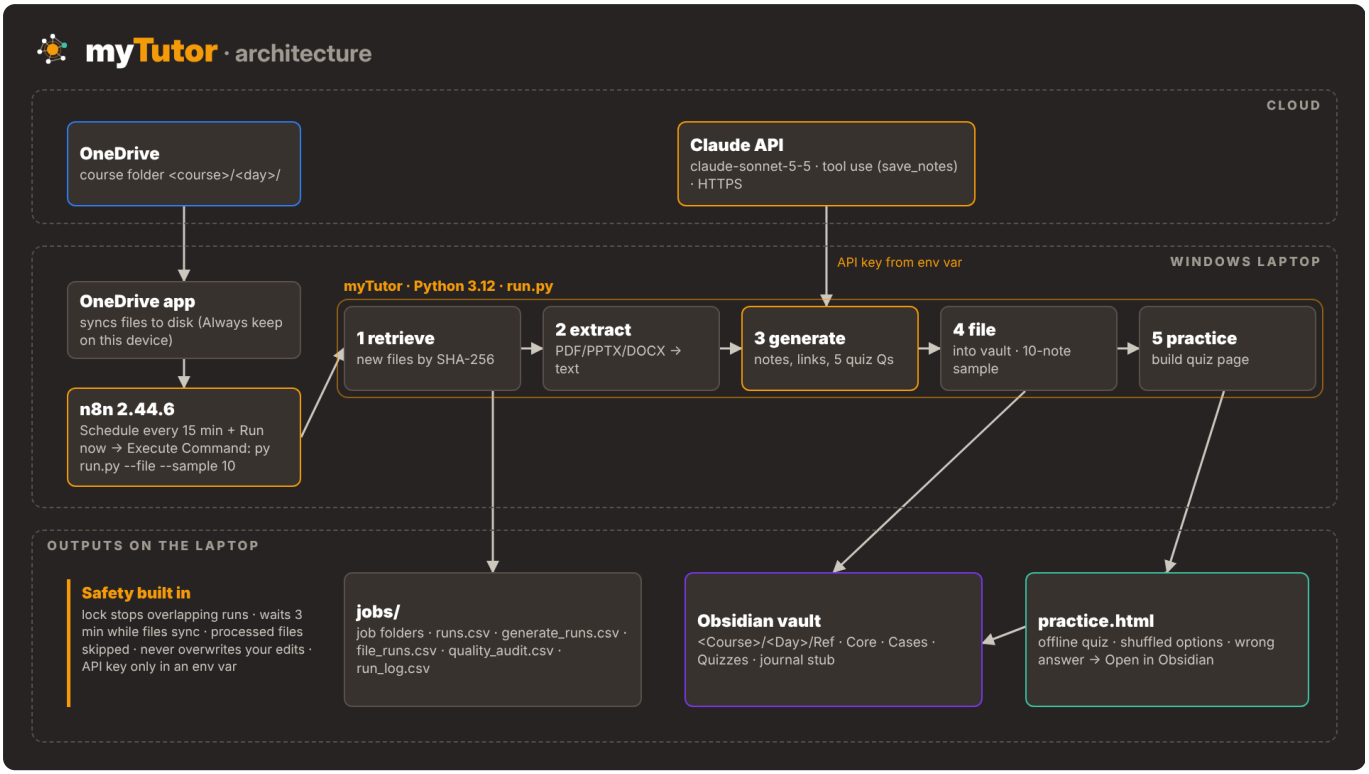
1. Overview
2. Architecture
3. Prerequisites
4. Technology stack and versions
5. Installation and setup
6. Configuration reference
7. Running and commands
8. Data, logs and folders
9. Testing
10. Security and responsible use
11. Troubleshooting
12. Known limits and roadmap

# 1. Overview

myTutor turns course files saved in OneDrive (lecture slides, readings, cases) into linked Obsidian notes and an offline practice quiz. n8n checks the course folder every 15 minutes and runs a Python pipeline that calls the Claude API once per source file.

| For each source file | myTutor produces |
|---|---|
| Reference note | Summary, key points, link back to the original file |
| Core notes | One note per key concept, linked to notes already in the vault |
| Case notes | One per real organisation discussed |
| Quiz | 5 multiple-choice questions; plus a 20-question practice set per teaching day |

# 2. Architecture




## Pipeline stages (run.py)

| Stage | What it does | Writes |
|---|---|---|
| 1 retrieve.py | Finds new files in `<course>/<day>/`, skips files already seen (SHA-256), copies them into a dated job folder | `source/`, `manifest.json`, `index.json`, `runs.csv` |
| 2 extract.py | PDF, PPTX and DOCX to clean text; counts words and pages; guesses the kind (slides, reading, case) | `text/*.md`, `extract.json` |

| | | |
|---|---|---|
| 3 generate.py | Calls Claude once per source with the vault's existing note titles; tool use returns notes and quiz as structured data; builds the day's 20-question set | `notes/` , `quiz/*.json` , `generate.json` , `generate_runs.csv` |
| 4 file.py | Copies notes into the Obsidian vault; never overwrites a note you edited (SHA check); picks a random 10-note spot-check sample | vault notes, `filed.json` , `file_runs.csv` , `quality_audit.csv` |
| 5 practice.py | Builds one offline HTML practice page from all saved quiz sets; options are shuffled on screen | `practice.html` |

> **Orchestration:** n8n's workflow has two triggers (a 15-minute schedule and a manual *Run now*) feeding one Execute Command node, then a Code node that reads the `SUMMARY {json}` line printed by run.py. Windows Task Scheduler running `run_mytutor.bat` is the fall-back.

## 3. Prerequisites

| You need | Why | Check |
|---|---|---|
| Windows 10/11 PC | Commands, .bat files and paths are Windows-style | — |
| Python 3.11 or later | Runs the pipeline (built and used on 3.12) | `py --version` |
| Node.js 24 or later | n8n 2.x refuses to run on Node 20 | `node -v` |
| n8n (npm) | Schedules and runs myTutor | `n8n --version` |
| Anthropic API key | Claude writes the notes; billed per token | Anthropic console |
| OneDrive app, signed in | Syncs the course folder to disk | Green ✓ on files |
| Obsidian, with a vault | Where notes are filed and read | Vault opens |
| Git (optional) | To clone the repository | `git --version` |

> **Skills assumed:** using Command Prompt, editing a JSON file, and installing desktop apps. No programming is needed to run myTutor; Python is needed only to change it.

## 4. Technology stack and versions

Versions in use for the experiment, checked 28 October 2026. "Minimum" is what the code requires.

| Component | Version used | Minimum | Role |
|---|---|---|---|
| Windows | 11 | 10 | Host OS |
| Python | 3.12.2 | 3.11 | Pipeline runtime |
| Node.js | 24.22.1 | 24 | Runtime for n8n |
| n8n | 2.44.6 | 2.x | Scheduler and orchestration |
| Claude model | claude-sonnet-5-5 | — | Note and quiz generation (tool use) |
| anthropic (Python SDK) | 1.15.0 | 1.0 | Claude API client |

| | | | |
|---|---|---|---|
| pymupdf | 1.28.2 | 1.24 | PDF text extraction |
| python-pptx | 1.0.2 | 0.6.23 | PowerPoint extraction |
| python-docx | 1.2.0 | 1.1 | Word extraction |
| msal, requests | optional | 1.28, 2.31 | Only for `"mode": "graph"` (OneDrive download via Microsoft Graph) |
| Obsidian | see Help › About | any recent | Note vault and graph view |
| Practice page | HTML + JavaScript, no libraries | any modern browser | Offline quiz |

Exact library versions installed on a machine: `py -m pip show anthropic pymupdf python-pptx python-docx`.

### Cost reference

Prices are set in `config.json` (`price_per_mtok`: input US$2, output US$10 per million tokens) and logged per source. Measured: about US$0.10 per file; 5 Day 9 files cost US$0.48; Day 1 (10 files, 34,255 words) cost US$1.01.

## 5. Installation and setup

### 5.1 Install the prerequisites

```
winget install Python.Python.3.12        (or python.org; tick "Add to PATH")
winget install OpenJS.NodeJS             (Node 24 or later)
npm install -g n8n
```

### 5.2 Get the code and libraries

```
cd %USERPROFILE%
git clone https://github.com/henrynawaung/mytutor.git
cd mytutor
py -m pip install -r requirements.txt
py -m unittest discover tests            (expect: OK, 67 tests)
```

### 5.3 Set the API key (environment variable only)

```
setx ANTHROPIC_API_KEY "your-key-here"
```

Open a new Command Prompt so the key is visible. Set a monthly spending limit in the Anthropic console.

### 5.4 Create config.json

Copy `config.example.json` to `config.json` and set the paths (section 6).

### 5.5 Prepare OneDrive

Create a folder per course with a sub-folder per teaching day, for example `MyTutor\MBAM601_...\Day 9\`. Right-click the course folder and choose **Always keep on this device**, so files are on disk rather than cloud-only.

### 5.6 First run by hand

```
py run.py --dry-run                    (which day folders would run)
py run.py --file --sample 10           (process, file into Obsidian, pick a sample)
py practice.py --open                  (open the quiz page)
```

### 5.7 Automate with n8n

1. Run `start-n8n.bat`. It sets `NODES_EXCLUDE=[]`, because n8n 2.x disables the Execute Command node by default.
2. Open `http://localhost:5678`, then **Import from file** › `n8n_mytutor_workflow.json`.
3. Check the Run myTutor command path, click **Execute workflow** once, then **Publish** so the schedule runs.
4. Fall-back: Task Scheduler running `run_mytutor.bat` every 15 minutes.

> The laptop must be on, n8n running and OneDrive synced for scheduled runs. Runs started by the schedule show without a flask icon in n8n's Executions list.

## 6. Configuration reference (config.json)

| Key | Default | Meaning |
|---|---|---|
| `mode` | synced | `synced` reads the OneDrive folder on disk; `graph` downloads via Microsoft Graph (needs an app registration) |
| `synced_inbox` | — | Folder holding `<course>/<day>/` sub-folders |
| `jobs_root` | ~/mytutor/jobs | Job folders, logs and `practice.html` |
| `vault_path` | — | Obsidian vault folder |
| `obsidian_vault_name` | vault folder name | Used in *Open in Obsidian* links |
| `term`, `term_label` | — | Term tags, e.g. T3 / 26T3; used to find course folders in the vault |
| `vault_folders` | Ref, Core, Cases, Quizzes, Theme | Sub-folder names inside each course/day |
| `model` | claude-sonnet-5-5 | Claude model |
| `price_per_mtok` | 2.0 / 10.0 | Input / output price per million tokens, for cost logging |
| `max_output_tokens` | 12000 | Cap per Claude call; a cut-off response is reported as an error |
| `extensions` | .pdf .pptx .docx .txt .md | File types read; others (images, .zip, .rar) are ignored |
| `max_file_mb` | 100 | Larger files are skipped |
| `settle_minutes` | 3 | A day folder changed more recently than this waits for the next run |
| `practice_questions` | 20 | Size of each day's practice set |

## 7. Running and commands

| Command | What it does |
|---|---|
| `py run.py --file --sample 10` | Full run: all stages, file into vault, 10-note sample (what n8n runs) |
| `py run.py --dry-run` | List day folders that would be processed |
| `py generate.py --day "Day 9" --estimate` | Token and cost estimate only |
| `py generate.py --day "Day 9" --only "Kowhai" --force` | Regenerate one source |
| `py generate.py --day "Day 10" --rebuild-set` | Rebuild the day's 20-question set (after adding files later) |
| `py file.py --day "Day 9" --dry-run` | Show what filing would change |
| `py file.py --day "Day 9" --sample 10 --no-file` | Pick a new spot-check sample only |
| `py practice.py --open` | Rebuild and open the practice page |

## 8. Data, logs and folders

```
mytutor/
  run.py retrieve.py extract.py generate.py file.py practice.py jobs.py
  config.json            your settings (not in Git)
  jobs/
    index.json           SHA-256 of every file already retrieved
    runs.csv             one row per retrieve
    generate_runs.csv    per source: tokens, cost, seconds, notes, links to existing notes
    file_runs.csv        per filing: filed, updated, unchanged, kept as you edited
    quality_audit.csv    10 sampled notes per run, five structural checks
    run_log.csv          one row per run.py run (incl. scheduled checks)
    practice.html        offline quiz page
    <date>_<course>-day-N/
      source/  text/  notes/  quiz/  manifest.json  extract.json  generate.json
```

Vault layout: `<Course folder>/<Day N>/{Ref, Core, Cases, Quizzes}` plus a journal stub left for the student.

### KPIs come from these logs

| KPI | Source |
|---|---|
| Time per source | `generate_runs.csv` seconds + timed review |
| Note quality | `quality_audit.csv` five checks |
| Links per note | `generate_runs.csv` links_to_existing ÷ notes |
| Recall | Practice page score, copied to the log |

## 9. Testing

```
py -m unittest discover tests
```

67 unit tests cover retrieval (duplicates, settling, size limits), extraction, response normalisation, note writing and linking, filing without overwriting edits, practice-set building and run orchestration. The answer-shuffle fix was retested with an automated browser on the real Day 9 set ( `docs/retest/` ): option B fell from 55% to 24% of correct answers over 200 runs, with marking unchanged.

## 10. Security and responsible use

> **API key:** only in the `ANTHROPIC_API_KEY` environment variable. Never in `config.json` , a text file or a commit. `.gitignore` blocks `config.json` , `key.txt` , `.env` and `jobs/` .

- **Course material** is sent to an AI service: process only files you are licensed to use and follow your institution's AI guidance. Course files are never committed to the repository.
- **Execute Command** in n8n runs shell commands; keep n8n on `localhost` and do not expose port 5678.
- **Graph mode** caches a token in `jobs/.token_cache.json` ; keep it private.
- **Accuracy:** notes pass structural checks automatically; factual accuracy needs a human spot-check (sample audit: 31 of 32 statements supported).

## 11. Troubleshooting

| Symptom | Fix |
|---|---|
| `py: can't open file ...\run.py` | Run `cd /d %USERPROFILE%\mytutor` first. |
| "Could not find platform independent libraries" | Harmless Python warning; ignore. |
| Day folder not processed | Files are cloud-only: set **Always keep on this device**; or a file changed in the last 3 minutes, so wait for the next run. |
| Read summary says "Nothing new" | Files already processed (by SHA). Use `--force` on generate.py to redo a source. |
| n8n: `EBADENGINE` or will not start | Upgrade Node.js to 24 or later. |
| n8n: Execute Command node missing / "?" | Start with `start-n8n.bat` (sets `NODES_EXCLUDE=[]` ). |
| n8n window blank, page "Cannot GET /" | Still starting; wait for "Editor is now accessible". Press Enter if the window is paused. |
| Schedule never runs | The workflow is not **Published**. |
| `ANTHROPIC_API_KEY is not set` | Run `setx` again and open a new window; restart n8n so it sees the key. |
| "Claude returned no usable notes" | Rerun that source with `--only "name" --force` ; blank templates or prompt files make poor sources. |
| Practice page missing new questions | `py generate.py --day "Day N" --rebuild-set` , then `py practice.py` . |

| | |
|---|---|
| Quiz notes flagged for no source link | Fixed in current version; rerun `file.py`. |

## 12. Known limits and roadmap

| Limit | Planned change |
|---|---|
| No OCR for scanned PDFs | Add OCR before extraction |
| Long books compress to few concepts | Split readings by section |
| No case notes for fictional teaching cases | Teaching-case rule in the prompt |
| Quiz distractors sometimes weak (AI review 72/100) | Prompt now asks for plausible distractors and a 20/50/30 recall/application/analytical mix; next: a validation step against the source before publishing |
| Scores kept in the browser only | Save results by concept to a file for learning tracking |
| Single user, single laptop | Shared vault and access control for team or company use |